1. Clinical Question & Core Hypothesis

Clinical Question

Immunologically 'hot' bladder tumors with visible lymphoid organization (high TLS/TIL) do not always benefit from immunotherapy—why do some TLS-rich tumors fail to respond?

Working Hypothesis

TLS-rich bladder tumors fail to mount productive anti-tumor immunity when high TGFB1 signaling and tumor SDC2 promoter hypermethylation cooperate to generate a fibrotic, immune-excluded niche that sequesters lymphoid aggregates and impairs effective antigen presentation. Consequently, TLS-high tumors with concurrent high TGFB1 expression and SDC2 methylation will show reduced response rates to immune checkpoint blockade compared with TLS-high tumors lacking these molecular features. The mechanism is supported by three interconnected axes: (i) TGFB1/SMAD-driven fibrosis and ECM remodeling that physically and chemically barrier cytotoxic T cells from TLS zones; (ii) SDC2 promoter methylation altering proteoglycan-mediated cell–cell signaling and stromal–epithelial communication, leading to diminished dendritic cell activation and decreased MHC-I–restricted antigen presentation; and (iii) modulation by secondary modifiers (TFPI2, MSH6, GREM1) that influence mutational load, neoantigen availability, and stromal signaling, thereby shaping the magnitude and quality of TLS-driven immunity. This yields two testable predictions: (P1) within TLS-rich bladder tumors, TGFB1high/SDC2methylated status associates with lower objective response rates to ICI and shorter progression-free survival; (P2) pharmacologic attenuation of TGFB1 signaling and/or reversal of SDC2 methylation will enhance TLS function and improve ICI efficacy in preclinical models and, prospectively, in treated patient cohorts.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TGFB1, SDC2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1 provides a mechanistic route to fibrosis, EMT, and immune exclusion; Core Path 2 (SDC2/TFPI2 methylation) because tumor epigenetic remodeling may alter epithelial/stromal interactions and antigenicity; Core Path 3 (Immunoscore/immune contexture) because the imaging/immune readout (TLS/TIL) is the starting phenotype that paradoxically sometimes lacks clinical benefit.

SDC2: SDC2/TFPI2 methylation) because tumor epigenetic remodeling may alter epithelial/stromal interactions and antigenicity; Core Path 3 (Immunoscore/immune contexture) because the imaging/immune readout (TLS/TIL) is the starting phenotype that paradoxically sometimes lacks clinical benefit.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	GREM1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective multi-omics analyses of bladder cancer patients treated with ICIs to correlate TLS metrics, TGFB1/SMAD readouts, and SDC2 promoter methylation with response and survival;

Prediction 2: spatial profiling (multiplex IHC, spatial transcriptomics) to map T cells relative to fibrotic ECM and SDC2-rich zones;

Prediction 3: in vitro models modulating TGFB1 signaling and SDC2 methylation (CRISPR/dCas9-based epigenetic editing and DNMT inhibitors) to assess effects on antigen presentation pathways;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TFPI2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.